PROGRESS LOG: WEEK 5-6

1. What I've done:

Data Collection:

- Completed full data collection for the 5km study area around the University of Strathclyde.
- Obtained and checked elevation data from SRTM from NASA (259,200 points at 30m resolution) through Open Topography API.
- Obtained daily rainfall data from the CEDA archive from the Met Office (MetUK-Grid) (2023 – 2025).

Exploratory Data Analysis:

- EDA was completed for all 6 datasets: SEPA PVA flood zones, OSM buildings, OSM roads, OSM water bodies, elevation and rainfall.
- Key finding: elevation is the strongest flood risk predictor, followed by distance to River Clyde which was added as a new feature after identifying it as more precise than general distance to water.

Feature Engineering:

- Elevation, dist_to_water, building_count, road_count, rainfall_annual, dist_to_clyde and flood_risk label are added to the feature matrix.
- Three-class flood risk label confirmed: High (17.3%), Medium (34.4%), Low (48.3%)

Ethics:

- Created Prepared Participant Information Sheet, Consent Form and complete Ethics Application answers.
- Awaiting formal CIS portal approval, supervisor has approved all documents through email.

2. Issues or barriers:

- Spatial variation across 5km study area was considered to be negligible with NASA GPM rainfall data, and subsequently replaced with Met Office HadUK-Grid daily data, which has meaningful spatial variation.

3. Next two weeks planning:

Technical:

- Train baseline Random Forest Classifier, and test with F1 score and AUC-ROC
- Run SHAP analysis to find the most important predictors of flood risk.
- Start creating Stream lit dashboard (Version A and Version B)

Ethics and Recruitment:

- Write dissertation chapters in addition to technical work.